B.TECH · LABORATORY DEMONSTRATION

Fiber Optic Splicing, OTDR Testing & Link Loss Measurement

Optical Fiber Communication Laboratory

Student name	*Your student name here*
Roll number	*Your roll number here*
Department	*Your department here*
College	*Your college here*
Faculty	*Your faculty here*

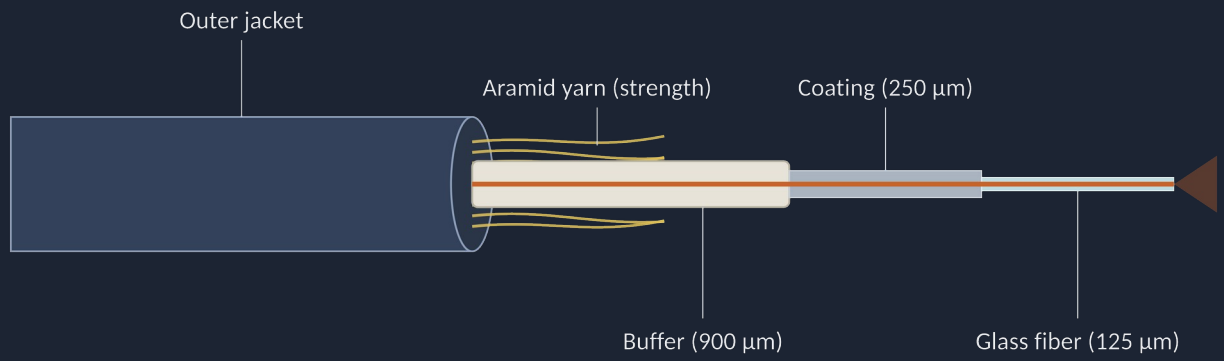


Illustration — layers not to scale

What We Will Learn

A → **B** → **C**

Experiment A

Fiber optic splicing

Join two optical fibers so light can pass from one to the other with very little loss.

Experiment B

Testing fiber with an OTDR

Send light pulses into the fiber and read the returned light to find its length, losses and faults.

Experiment C

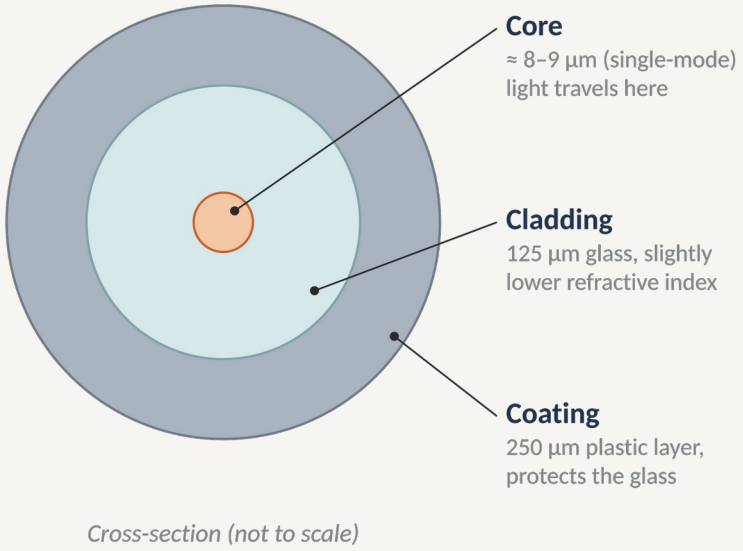
Link loss using the OTDR

Read the total loss of the fiber link in dB and work out the average loss per kilometre.

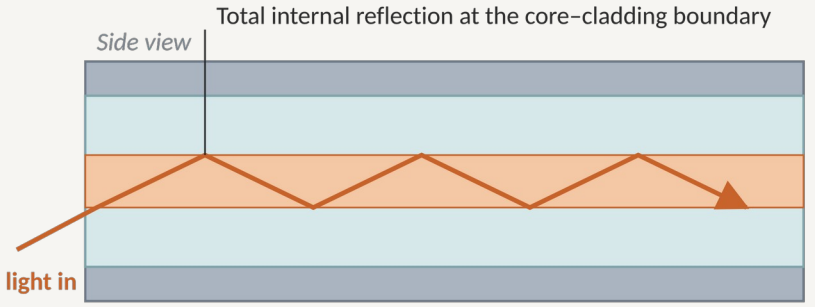
How the experiments connect

A splice is only useful if we can prove it is good. So after joining fibers (A), we use the OTDR to look along the fiber and see where the light is lost (B). Then we use the same trace to read the total loss of the link and compare it with the fiber's length (C).

Understanding Optical Fiber



Cross-section (not to scale)



Core index n_1 is a little higher than cladding index n_2. Light hitting the boundary at a shallow angle bounces back into the core, so it stays trapped and travels long distances.

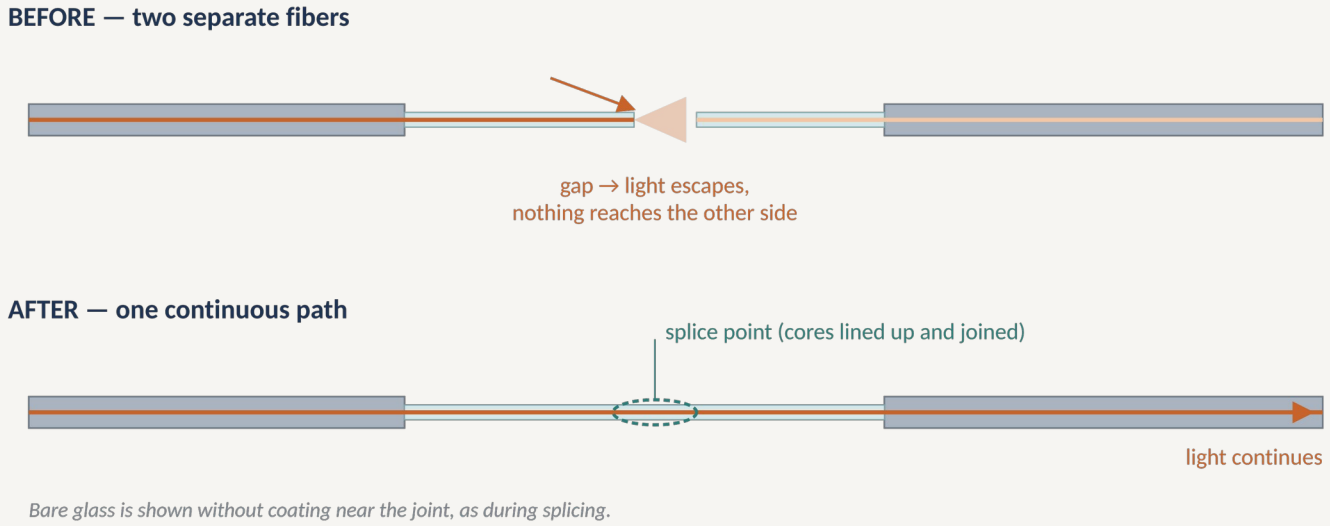
Why use fiber for communication?

- Very low loss, so signals go many km without a repeater
- Huge bandwidth — lots of data at once
- Not affected by electrical noise or lightning
- Thin, light and hard to tap

What is it? A very thin strand of pure glass (about the thickness of a hair) that carries information as light.

How does light stay inside? The core's glass bends light a little more than the cladding's. Light that hits the edge at a shallow angle reflects back in, again and again.

What Is Fiber Optic Splicing?



Bare glass is shown without coating near the joint, as during splicing.

The core is only about 9 µm wide
— roughly a tenth of a human hair. If the two cores are off by even a few micrometres, part of the light misses the second core and is lost.

Meaning

Splicing is joining two optical fibers end-to-end so light passes from one into the other.

Why join fibers?

Cables come in fixed lengths, cables get cut or damaged, and pigtails must be joined to the main cable.

Where is it used?

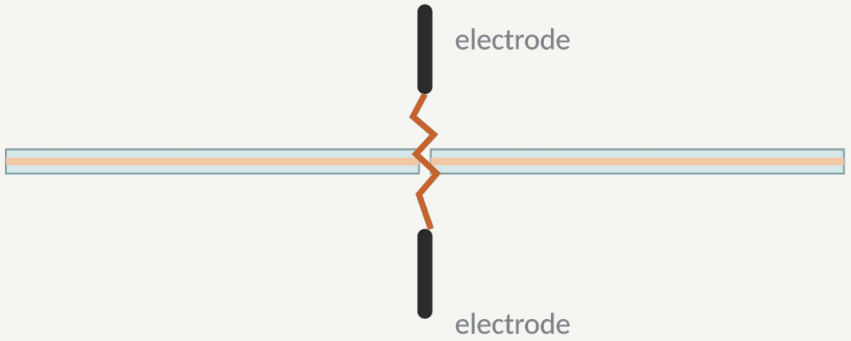
Long-distance links, FTTH (fiber to the home), joint closures, patch panels and repair work.

Why alignment matters

Light only travels in the tiny core. Good alignment and clean, flat ends give low loss.

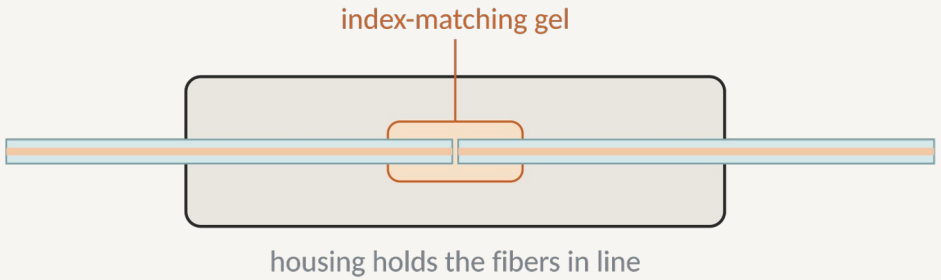
Types of Splicing

Fusion splice



An electric arc melts the glass ends together.

Mechanical splice

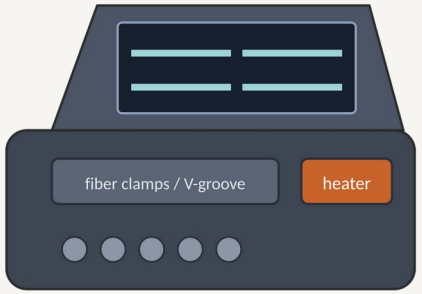


Fiber ends are pushed together and clamped.

	Fusion splicing	Mechanical splicing
How it joins	Glass ends are melted together by an electric arc	Fiber ends are held face-to-face in a small housing with gel
Typical loss	Very low — often below 0.1 dB	Higher — often a few tenths of a dB
Back-reflection	Very low	Low, but higher than fusion (depends on gel)
Tools & cost	Fusion splicer — costly machine	Simple tools — low cost
Best for	Backbone and long links, many splices	Quick repairs and a few splices

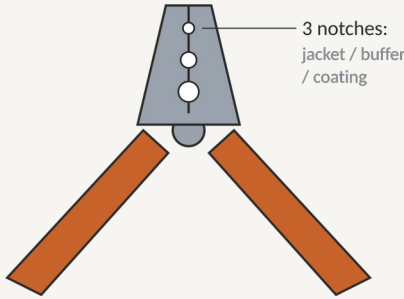
Loss values are typical figures from fiber-optic training guides (e.g. FOA). Real loss depends on the fibers, the tools and the person splicing.

Equipment Required for Fusion Splicing



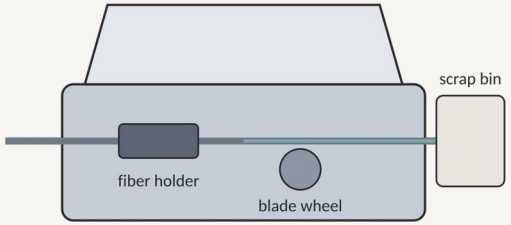
Fusion splicer

Aligns the two fibers with cameras and motors, then fuses them with an electric arc.



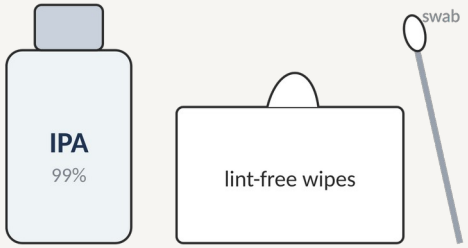
Fiber stripper

Removes the jacket, buffer and 250 µm coating without scratching the glass.



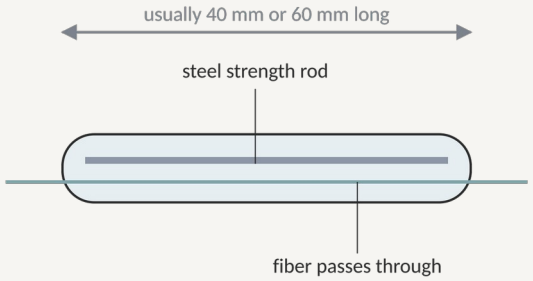
Precision cleaver

Cuts the bare fiber so the end is flat and at 90° to its length.



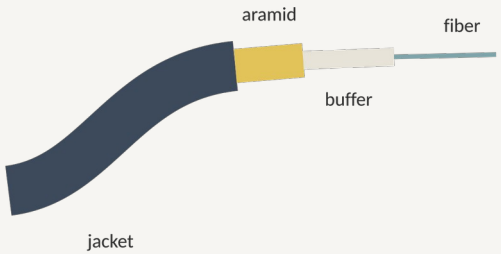
Cleaning kit

99% isopropyl alcohol and lint-free wipes remove dust and oil from the glass.



Heat-shrink sleeve

A tube with a steel rod that shrinks over the joint to protect it.



Optical fiber cable

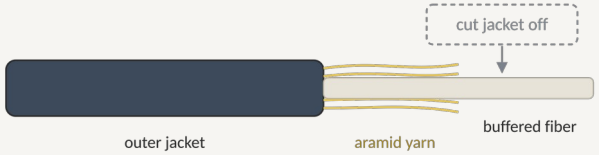
The fiber to be joined; check its type before you start.

Fusion Splicing (1/4): Preparing the Fiber



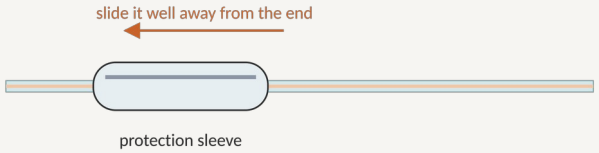
1 Prepare the cable

Check the fiber type (single-mode or multimode) printed on the cable. Cut a clean, straight end and mark how much jacket to remove. Work on a clean, flat table.



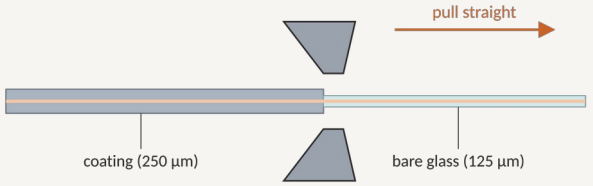
2 Remove the outer jacket

Score the jacket with a jacket stripper and slide it off. Cut the aramid yarn with sharp scissors — never with the cleaver.



3 Slide on the protection sleeve

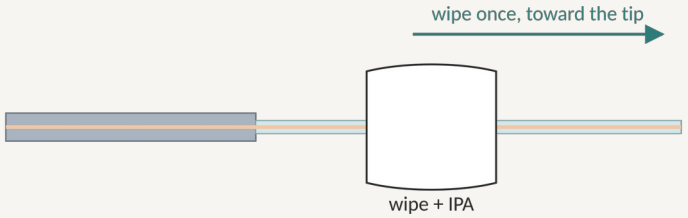
Put the heat-shrink sleeve onto ONE fiber now and push it well away from the end. If you forget, you will have to break the splice and start again.

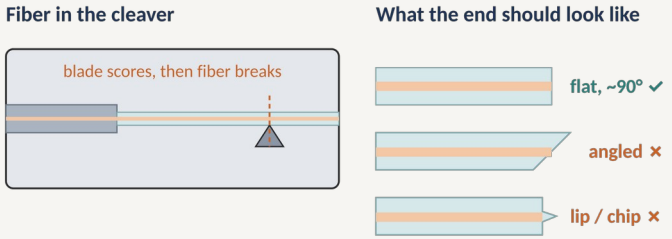


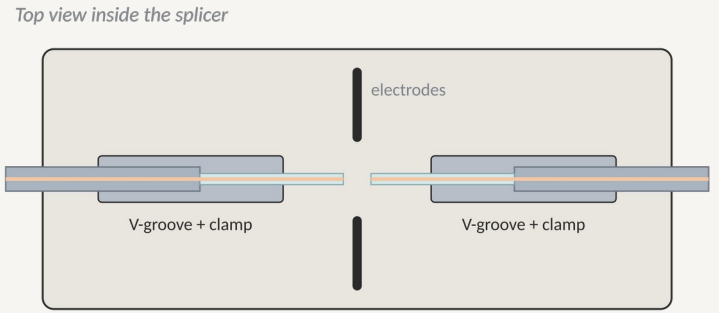
4 Strip the fiber coating

Use the smallest notch of the stripper. Hold it square and pull in one smooth stroke to expose the 125 µm glass (about 3–4 cm, as the cleaver guide says).

Fusion Splicing (2/4): Clean, Cleave, Load







5 Clean the bare fiber

Wipe the glass once, towards the tip, with a lint-free wipe and 99% isopropyl alcohol. Do not touch it again after cleaning.

6 Cleave both fiber ends

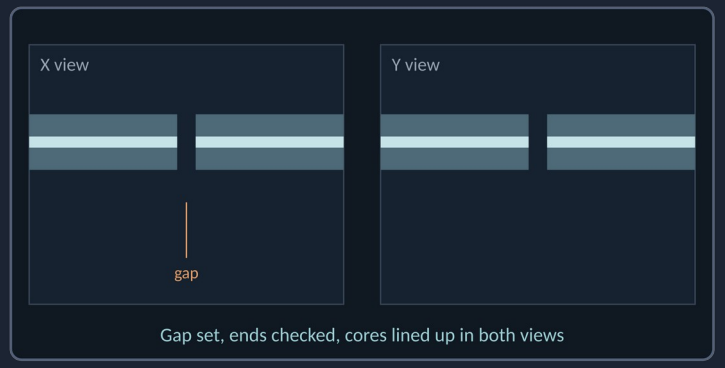
Set the length on the cleaver scale, place the fiber and close the lid. A good cleave is flat and square (usually within 1°). Drop the offcut into the scrap bin at once.

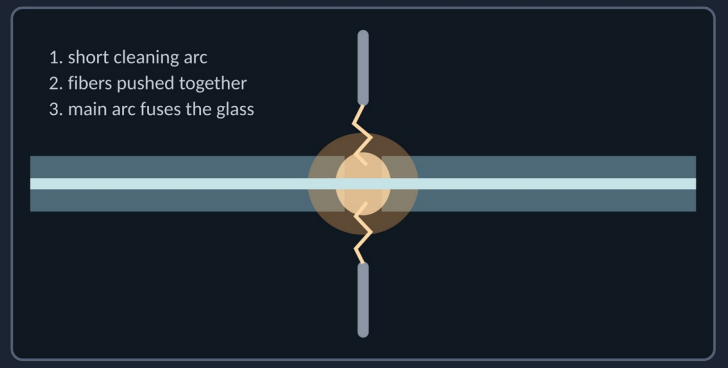
7 Place fibers in the splicer

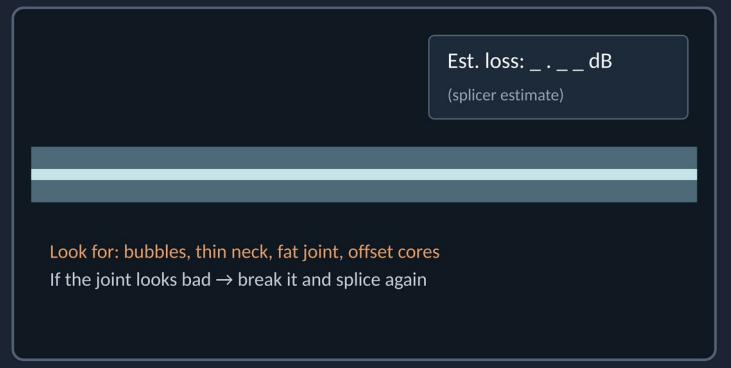
Lay each fiber in its V-groove with the tip just short of the electrodes. Close the clamps and the wind cover gently. The clean tip must not touch anything.

Safety: bare fiber offcuts are tiny glass needles. They can enter the skin or eyes. Pick them up with tape and put them in the scrap bin — never on the floor or table.

Fusion Splicing (3/4): Inside the Splicer







8 Align both fiber cores

The splicer's cameras look at the fibers from two directions (X and Y). It checks the end faces and the gap, then moves the fibers until the cores line up.

9 Fuse with the electric arc

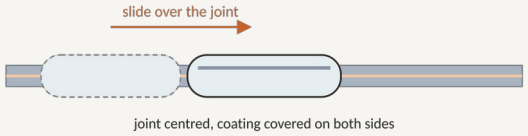
A short arc first cleans the ends. The fibers are pushed together and the main arc melts the glass into one piece. This takes only a few seconds.

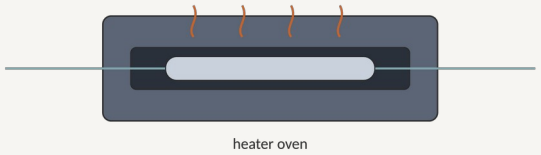
10 Inspect and check estimated loss

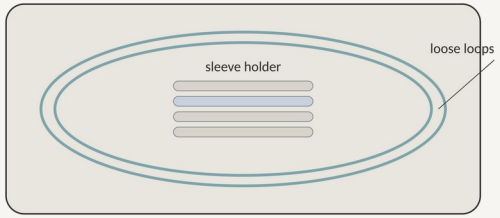
Look at the joint on screen for bubbles, thin or fat spots, or offset cores. The loss shown is the splicer's estimate from the picture — not a real measurement.

Screen drawings are illustrations of a typical splicer display, not screenshots from our machine. We did not record a splice-loss value, so none is shown.

Fusion Splicing (4/4): Protect and Test









11 Move the sleeve over the joint

Open the clamps, lift the fiber with gentle tension, and slide the sleeve so the joint sits in the middle.

12 Heat-shrink the sleeve

Put the sleeve in the heater and run the heat cycle. It shrinks and grips the joint. Let it cool before handling.

13 Arrange in the splice tray

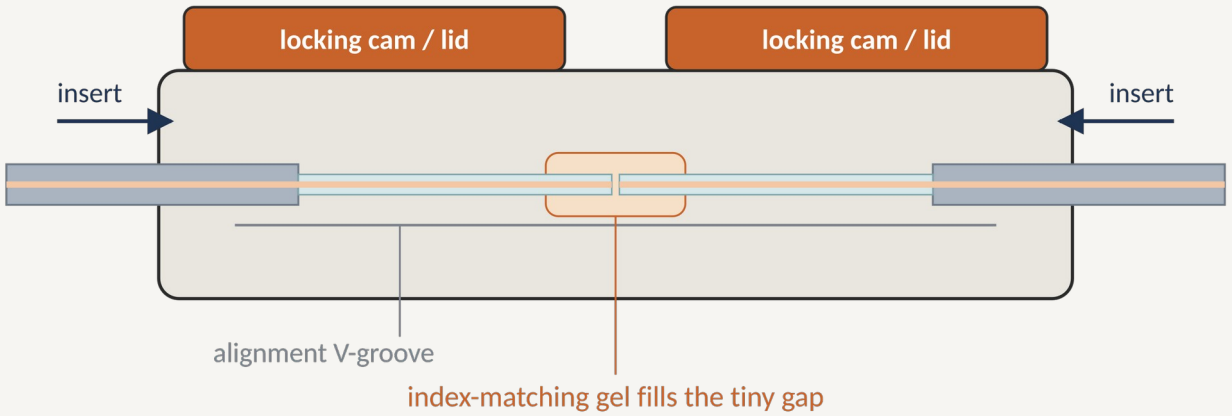
Press the sleeve into the tray holder and coil the spare fiber in large loops. Never bend it sharply.

14 Test the connection

Test with an OTDR or a light source and power meter. Record the actual measured loss — not the splicer's estimate.

Quick order check: prepare → jacket off → sleeve on → strip → clean → cleave → load → align → fuse → inspect → sleeve over → heat → tray → test

Mechanical Splicing

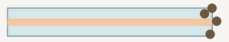
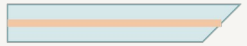
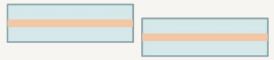
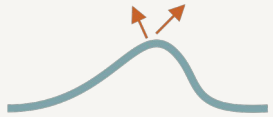
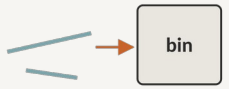


Mechanical splice — cut-away view (illustration)

No arc and no machine. The two fibers are pushed into a small housing. A V-groove lines them up, and a gel with nearly the same refractive index as glass fills the tiny gap so less light is reflected or lost.

1	**Strip**	Remove coating to the length given by the splice maker.
2	**Clean**	Wipe the bare glass with alcohol and a lint-free wipe.
3	**Cleave**	Make a flat, square end.
4	**Insert**	Push the first fiber into one side, up to the middle.
5	**Align**	Push in the second fiber until the ends meet; a slight bow shows contact.
6	**Lock**	Close the cam or lid to clamp both fibers.
7	**Test**	Measure the loss with an OTDR or power meter.

Splicing Precautions and Common Mistakes

	PROBLEM		WHAT TO DO
 dust on the end face	**Dirty fiber ends** Dust or skin oil burns in the arc and causes bubbles and high loss.	→	Clean with 99% IPA just before cleaving. Do not touch bare glass.
 angled end face	**Poor cleaving** Angled, chipped or lipped ends do not fuse well.	→	Keep the blade clean, rotate it when worn, and re-cleave any bad end.
 cores offset	**Incorrect alignment** If cores are offset, part of the light misses the next core.	→	Seat fibers properly in the V-grooves; clean the grooves; let the splicer align.
 light leaks at a tight bend	**Sharp fiber bends** Light leaks out at tight bends (worse at 1550 nm).	→	Keep large loops in the tray and follow the bend-radius guides.
 glass slivers → sealed bin	**Unsafe fiber fragments** Glass slivers can enter skin and eyes.	→	Wear safety glasses, use a scrap bin, don't eat or drink at the bench.

What Is an OTDR?

OTDR stands for **Optical Time Domain Reflectometer**. It is used to find the length, losses and faults in an optical fiber.

Length

How long the fiber is, and where each event is (in km).

Loss

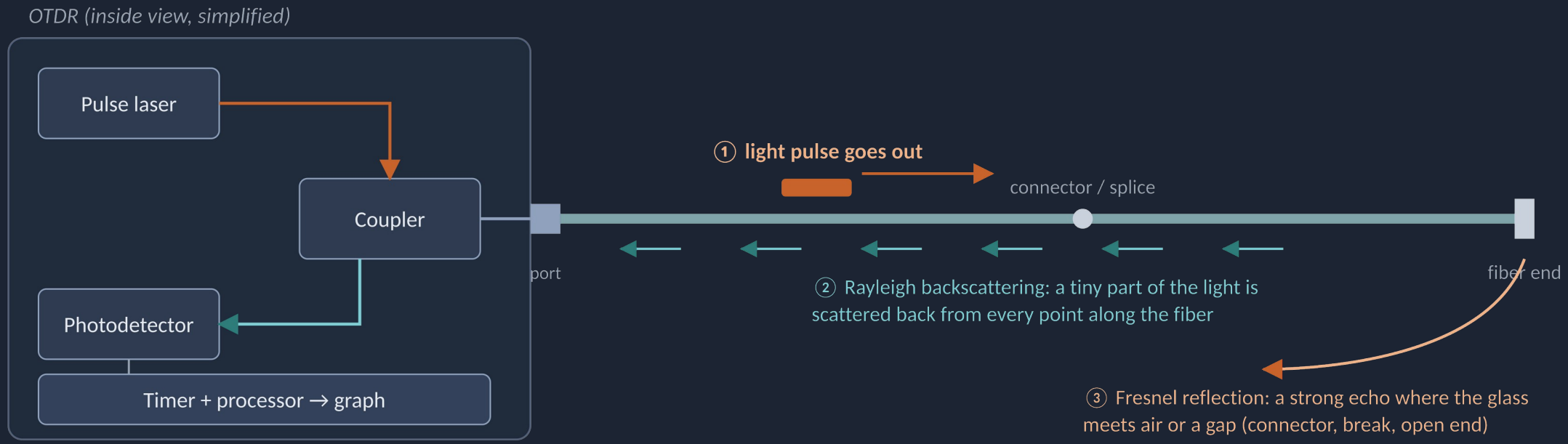
How much light is lost along the fiber and at each splice or connector (in dB).

Faults

Where a break, bad splice or sharp bend is, so a repair team knows where to dig.

Why it is so useful in practice: the OTDR needs access to only ONE end of the fiber. A power meter test needs equipment at both ends.

How an OTDR Works



④ Time of the echo → distance: $d = (c \times t) / (2\,n)$

c = speed of light in vacuum, t = round-trip time, n = group index of the fiber (≈ 1.47). Divide by 2 because light goes out and back.

1. Sends a short light pulse into the fiber
2. Some light is scattered or reflected back
3. The OTDR's detector receives the returned light
4. It measures how long the light takes to return
5. It works out distance and signal loss
6. It draws the result as a graph (trace)

Equipment Used in Our Experiment

FiberShot Pro Series OTDR — the instrument used in our lab. Both photos below were taken by us during the experiment.

LAB PHOTO

Photo 1 — OTDR test-settings screen

Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

LAB PHOTO

Photo 2 — OTDR trace and result screen

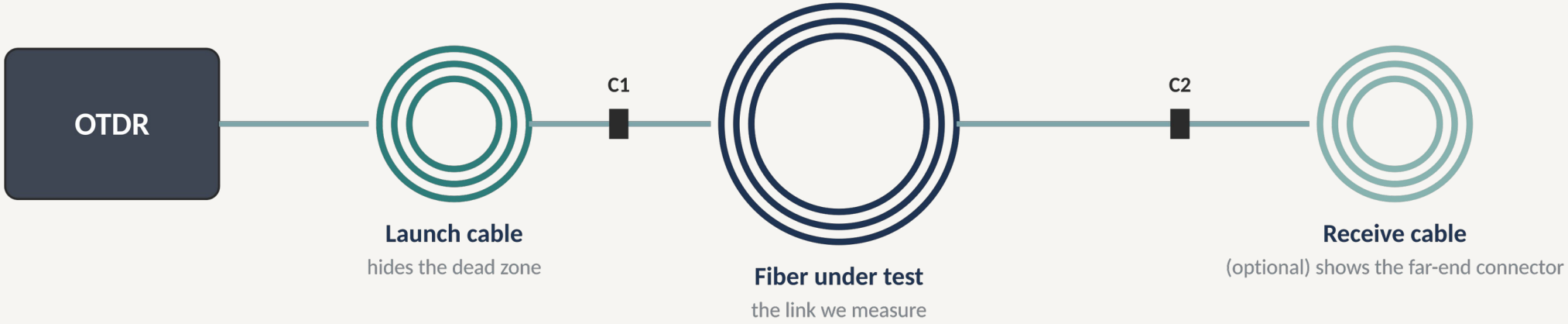
Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

Photo 1 · Test-setting menu: wavelength, range, pulse width, test time, resolution, launch/end cable and power mode.

Photo 2 · Trace with distance, total loss, attenuation slope, ORL and the PASS result.

Connecting the OTDR

Connection order



1. Switch on the instrument and let it start up.
2. Inspect and clean every connector end face.
3. Connect the launch cable to the OTDR port (if available).
4. Connect the fiber under test.
5. Check that all connections are firm and secure.
6. Make sure the fiber has no live signal on it (unless using a live-fiber OTDR setup).

Why launch and receive cables?

Right after the OTDR port there is a 'dead zone' where the strong reflection blinds the detector. A launch cable moves the first connector out of this zone so it can be measured. A receive cable does the same for the far-end connector.

⚠️ **Never look into the OTDR port or a fiber end — the laser light is invisible and can harm your eyes. Use an inspection scope.**

OTDR Test Settings We Used

Parameter	Our value	What it means
Test wavelength selection	**1310/1550 nm**	Wavelength(s) of laser light used for the test
Test range	**10 km**	Longest distance shown; set longer than the fiber
Pulse width	**50 ns**	Pulse length: short = fine detail, long = reaches farther
Test mode	**Manual**	We chose the settings ourselves (not Auto)
Test time	**50 seconds**	Averaging time: longer gives a smoother trace
Resolution	**Standard**	How finely the trace points are spaced
Launch cable setting	**None**	No launch cable length entered or removed
End cable setting	**None**	No receive (end) cable entered
Power mode	**Standard**	Instrument power option, left at default

Short pulse + long averaging = a clear, detailed trace for a short (few-km) fiber like ours.

LAB PHOTO

Photo 1 — test-settings screen

Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

Values copied exactly from our settings screen (Photo 1).

Performing the OTDR Test

1. Open the Test Setting menu.
2. Select the wavelength.
3. Set the measurement range.
4. Select the pulse width.
5. Set the test duration.
6. Select resolution and power mode.
7. Return to the main test screen.
8. **Press AVG to start the averaged test.**
9. Wait for the test to finish.
10. Observe the trace and the event table.



Why 'average'?

The returned light is very weak and noisy. The OTDR repeats the pulse many times during the test time (50 s for us) and averages the results. Noise cancels out and the trace becomes clean.

Understanding the OTDR Graph

LAB PHOTO

Photo 2 — our OTDR trace (1310 nm)

Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

- **X-axis**
  Distance along the fiber, in kilometres.
- **Y-axis**
  Level of the returned signal, in dB.
- **Sloping straight line**
  Normal fiber attenuation — the light weakens a little every km.
- **Sudden downward step**
  A loss event: possibly a splice, bend or other loss point.
- **Upward spike**
  A reflection: usually a connector or the fiber end.
- **Sharp drop to noise**
  End of the measured fiber link.

We only mark events that the OTDR actually shows on our screen and event table. We do not guess faults that are not on the trace.

Our OTDR Observations

Parameter	Observed value
Test wavelength	**1310 nm**
Pulse width	**50 ns**
Fiber distance	**4.822 km**
Total loss	**1.550 dB**
Displayed attenuation slope	**0.317 dB/km**
Optical Return Loss (ORL)	**29.37 dB**
Instrument result	**PASS**

All values read directly from the result screen (Photo 2). Nothing here is calculated.

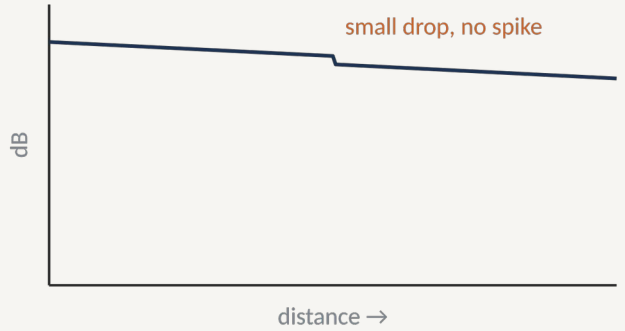
What PASS means: the measured results were within the thresholds set in the instrument. It is not proof that the link meets every possible performance requirement.

LAB PHOTO

Photo 2 — result screen

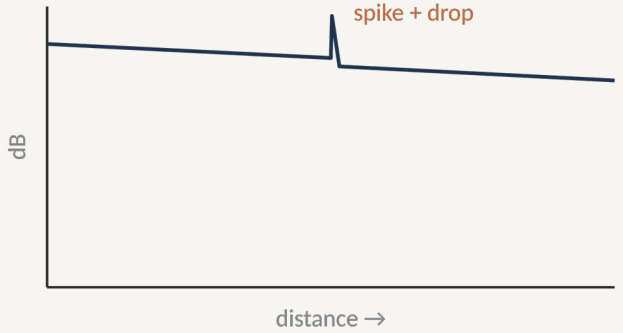
Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

Common OTDR Events



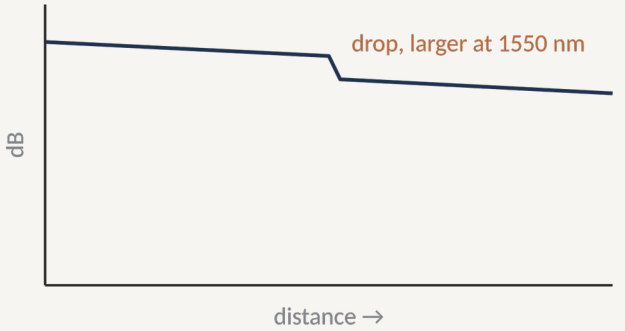
Fusion splice

Small step down. No spike, because a fused joint hardly reflects.



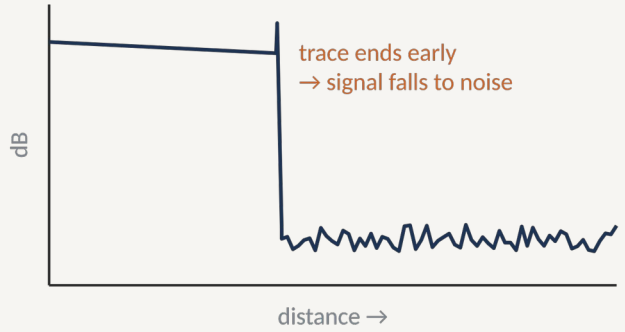
Connector

Spike (reflection from the glass gap) followed by a step (loss).



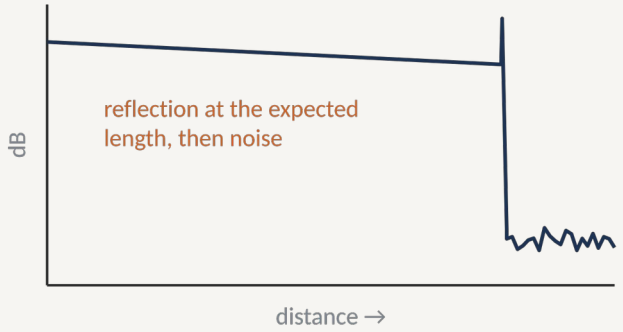
Sharp bend

Step down with no spike. The loss is bigger at 1550 nm than at 1310 nm.



Fiber break

Trace ends earlier than the known length and falls to noise.



Fiber end

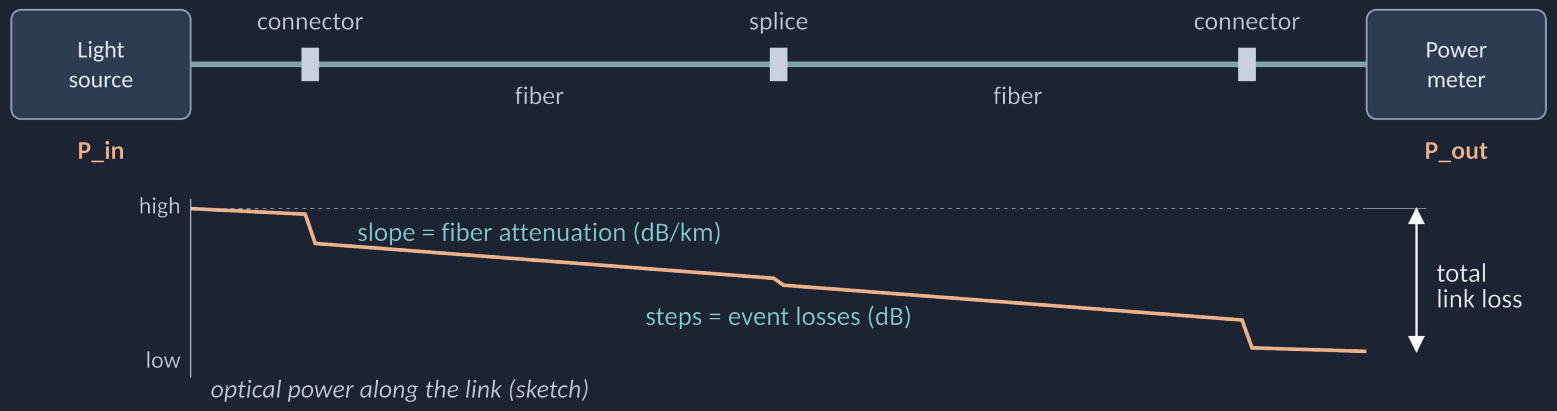
Reflection at the expected length, then only noise.

Reflective events (connector, end, break with a gap) show a spike.

Non-reflective events (fusion splice, bend) show only a step.

What Is Optical Link Loss?

Link loss is the total reduction in optical signal power as light travels through a fiber link.



Total link loss

Unit: dB. Everything lost from one end to the other — fiber + splices + connectors.

Fiber attenuation

Unit: dB/km. How much the fiber itself loses per kilometre. It is the slope of the trace.

Example (not our data): 10 km of fiber at 0.35 dB/km loses 3.5 dB in the glass alone. Add each connector and splice loss to get the total link loss.

Procedure to Measure Link Loss Using OTDR

01 Clean and inspect the fiber connectors.

02 Connect the OTDR to the fiber link.

03 Use launch / receive cables when required.

04 Open the test settings.

05 Select wavelength, range and pulse width.

06 Start the averaged (AVG) measurement.

07 Observe the trace.

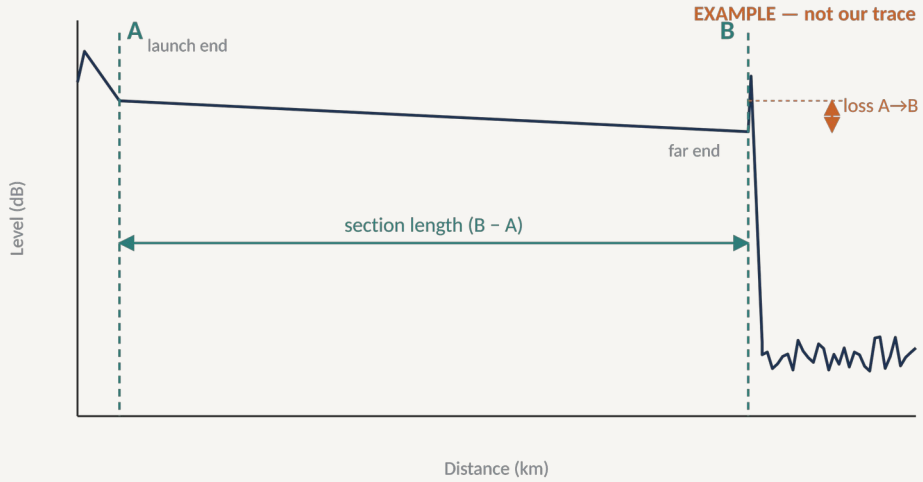
08 Find the start and the end of the link.

09 Use the event table or the A–B loss cursors.

10 Read the section loss in dB.

11 Record the fiber length and the loss.

12 Repeat at another wavelength if required.



Example: cursors A and B placed at the start and end of the fiber section.

Important: the OTDR estimates loss from the backscattered light. It does not directly measure the power sent in at one end and received at the other end — that is a power-meter test.

Link Loss Formulas

$$\text{Link Loss (dB)} = P_{in}\ \text{(dBm)} - P_{out}\ \text{(dBm)}$$

Basic end-to-end insertion loss — used with a light source and an optical power meter.

With an OTDR, we do not use this formula. The link or section loss (dB) is read from the event table or from the A–B loss-measurement cursors.

$$\text{Average Attenuation (dB/km)} = \textbf{Total Loss (dB)} \div \textbf{Fiber Length (km)}$$

Gives the average loss for every kilometre of the link.

Because dBm − dBm gives dB, the loss comes out in dB, not dBm.

Terms

P_{in}	Optical power sent into the link, in dBm.
P_{out}	Optical power received at the far end, in dBm.
dBm	Power compared with 1 mW (0 dBm = 1 mW).
dB	A ratio — how much power is lost.
Total Loss	Loss of the measured section, in dB.
Fiber Length	Length of that section, in km.

Calculation Using Our Readings

GIVEN (measured)

Fiber length, L = **4.822 km**

Total loss = **1.550 dB**

CALCULATION

Average attenuation = Total loss / L

= 1.550 dB / 4.822 km

≈ 0.3214 dB/km

0.3214

dB/km · calculated by us
(total loss ÷ length)

0.317

dB/km · attenuation slope
shown by the OTDR (measured)

Why are they slightly different?

- The OTDR's slope is a straight-line fit over a smooth part of the trace. It leaves out events such as splices and connectors.
- Total loss covers the whole section, so any event loss inside it is included.
- So 0.317 is NOT simply 1.550 ÷ 4.822 — the two numbers come from different methods.

Difference = 0.3214 − 0.317 ≈ 0.004 dB/km (about 1.4%). Check: 0.317 dB/km × 4.822 km ≈ 1.529 dB, which is about 0.02 dB less than the measured 1.550 dB.

Experiment Observation Report

OBSERVATION SHEET

Date: _____ Batch: _____

Experiment: Optical fiber testing and link loss measurement using OTDR

Equipment: FiberShot Pro Series OTDR, fiber under test, connector cleaning kit

Parameter	Value	Parameter	Value
Test wavelength	**1310 nm**	Measured fiber length	**4.822 km**
Test range	**10 km**	Total loss	**1.550 dB**
Pulse width	**50 ns**	Attenuation slope (displayed)	**0.317 dB/km**
Test time	**50 s**	Optical Return Loss	**29.37 dB**
Launch / end cable	**None / None**	Instrument result	**PASS**
Average attenuation (calculated)	**0.3214 dB/km**	Resolution / power mode	**Standard / Standard**

Conclusion: The OTDR measured a 4.822 km fiber with 1.550 dB total loss at 1310 nm (average ≈ 0.321 dB/km). The instrument showed PASS against its set thresholds.

Student's signature _____ Faculty's signature _____

Results and Discussion

4.822 km

fiber length · at 1310 nm

1.550 dB

total loss · at 1310 nm

PASS

instrument result · at 1310 nm

What the test showed

The OTDR found the fiber to be about 4.822 km long, with a total loss of 1.550 dB.

PASS

The instrument showed PASS. This only means the results were inside the thresholds set in the OTDR.

What we learned

The test showed how an OTDR inspects a link from one end: length, loss and events from a single trace.

Limitation

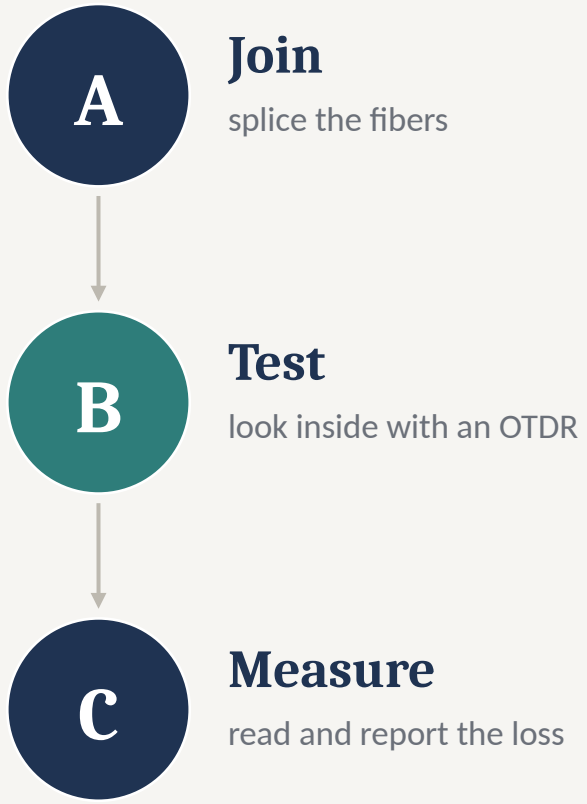
Launch and end cable settings were 'None'. So the loss of the first and last connectors cannot be measured accurately — they fall in the dead zone or at the end reflection.

Improvement

Repeat with launch and receive cables, and at 1550 nm, to compare results.

Conclusion

1. Fiber splicing joins two optical fibers so light can continue from one to the other.
2. Fusion splicing uses heat from an electric arc to make a permanent, very low-loss joint.
3. An OTDR measures fiber length and loss, and finds faults — from one end of the fiber.
4. Link loss can be estimated from OTDR trace measurements (event table or A–B cursors).
5. Proper cleaning, cleaving, alignment and testing give a good fiber connection.



References

Textbooks and standards

1. G. Keiser, Optical Fiber Communications, 5th ed., McGraw-Hill Education.
2. J. M. Senior, Optical Fiber Communications: Principles and Practice, 3rd ed., Pearson Education, 2009.
3. ITU-T Recommendation G.652 — Characteristics of a single-mode optical fibre and cable. itu.int
4. IEC 61746-1 — Calibration of optical time-domain reflectometers (OTDR), Part 1: single-mode fibres. iec.ch
5. IEC 60825-1 — Safety of laser products. iec.ch

Practical guides and documentation

6. The Fiber Optic Association (FOA) Online Reference Guide — sections on fusion splicing, mechanical splicing, and OTDR testing. thefoa.org/tech/ref/
7. ANSI/TIA-568.3-D — Optical Fiber Cabling and Components Standard (loss limits for splices and connectors).
8. User manual of the FiberShot Pro Series OTDR used in the lab (for menu names and settings).
9. User manuals of the fusion splicer and cleaver available in the lab (for strip and cleave lengths).

All diagrams in this presentation were drawn by us for this lab. The two OTDR photographs were taken during our experiment.

Thank You

Questions?